Miguel Esteban Pérez Ibarra
Escuela Superior de Física y Matemáticas
Instituto Politécnico Nacional, Mexico City, Mexico
supersluky@gmail.com

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The Editor
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Dear Editor,

I am pleased to submit the manuscript 'A Pythagoras Tree Grown from a Dodecahedron' for consideration as a Research Article in *The American Mathematical Monthly.*

Bosman's Pythagoras tree is a favorite picture of recreational mathematics, and its three-dimensional versions are built from cubes. The article describes a Pythagoras tree built instead from regular dodecahedra. It began as a piece of computer art, and the article tells how its recipe was recovered from the original files, why it never collides with itself, and how a classical fact about the six face axes of the dodecahedron, a 'Pythagorean theorem' for the dodecahedron, picks out a special tree in which every child rests flat on its parent, surface area is conserved from one generation to the next, and the limit set has dimension exactly two. It closes with a short counting argument that decides when the tree can grow from all twelve faces at once.

I believe the article suits the *Monthly* because it brings together ideas a broad readership knows, namely the Pythagorean theorem, the Platonic solids, symmetry groups, and self-similar fractals, around a concrete object that is easy to picture. The prerequisites are undergraduate linear algebra and a little group theory, and the figures carry much of the story. One proof is computer-assisted; it reduces an infinite question to finitely many linear programs, and all code and data are publicly available.

The manuscript is original, has not been published, and is not under consideration elsewhere. English is not my native language, and generative AI was used to help write parts of the text and to help produce the Blender renders; this is described in the declaration included in the manuscript. The figures are in color; color is needed only in the online version, and black-and-white print is acceptable. No funding was received, and I have no competing interests to declare.

Sincerely,

Miguel Esteban Pérez Ibarra